

A woman in a blue suit stands in a server room aisle, looking at a laptop on a stand. The room is filled with blue server racks on both sides, and the floor is a light-colored tile. The lighting is bright and even.

Building a Highly-available Web Server Cluster

Getting started with nginx.

*n*ginx (pronounced as 'engine x') is a powerful HTTP Web server/reverse proxy and IMAP/POP3 reverse proxy. According to a survey conducted in December 2008 by Netcraft, nginx has grown significantly and has surpassed lighttpd (also known as Lighty). Because of its small memory footprint and high scalability, nginx has found tremendous usage in virtual private servers (VPS).

A reverse proxy is a front end to one or more Web servers. All connections originating from the Internet and destined for Web servers (behind the reverse proxy) are routed through the proxy. Based on the configuration, the proxy may decide to serve the request itself or pass it either partially

or totally to one of the member Web servers behind it. In this manner, the reverse proxy presents a single interface for a set of servers, to the caller. A reverse proxy can thus work as a load balancer distributing the incoming load to member servers. It can also cache the contents being served.

The architecture: A layered approach

I typically use Debian or RHEL in my servers and try to stick with the packages available in the base distribution as far as possible.

For the purpose of this article, I will use four Debian Etch servers and will divide the entire set-up into two layers. The first layer (Layer 1) will have a pair of highly-available (using Heartbeat) *nginx* reverse proxy